Project 4: Password Generation/Management Tool

CS 428/528: Computer Security

Project Proposal Due Date : October 27, 2026
Project Due Date : November 9, 2026

Overview

In this project, you will work in small groups (**up to 2 students per group**) to design and implement a secure password generation tool inspired by a modern password manager. Each team must select an existing password manager and replicate the core password generation functionality, including configurable options and usability features.

Objective

The goal is to design a secure, configurable, and user-friendly password generator that reflects modern best practices.
Your tool should go beyond simply generating random strings and instead demonstrate understanding of:

- Entropy and randomness
- Secure random number generation
- Character set selection
- Length and complexity tradeoffs
- Usability vs. security balance

Reference Password Managers

Each team must choose one of the following (or request approval for another):

- LastPass, Bitwarden, 1Password, Dashlane, Keeper, NordPass, Proton Pass, RoboForm

Your implementation should replicate the generator behavior and configuration options of your selected manager.

Project Proposal (Required)

Each team must submit a short proposal (1 to 2 pages) that includes:

- The selected password manager
- Screenshots of its password generation interface
- Description of configurable options (length, symbols, digits, etc.)
- Initial design plan for replication
- Planned implementation language

Proposals must be approved before development begins.

Project Expectations

Your final tool should:

- Allow configurable password length
- Allow inclusion/exclusion of:
  - Uppercase letters
  - Lowercase letters
  - Numbers
  - Special characters
- Use a cryptographically secure random number generator
- Provide user-friendly interface

Bonus features may include:

- Passphrase generation mode
- Avoid ambiguous characters
- Copy-to-clipboard functionality
- Strength visualization

Use of AI

You are encouraged to use AI tools for brainstorming and design refinement. However:

- You must understand and explain your implementation.
- You must document AI usage.
- Secure randomness implementation must be verified by your team.

What to Submit

- Source code
- Project report (3 to 5 pages) describing:
  - Design choices
  - Randomness source
  - Comparison to original tool
- AI usage statement
- Final in-class presentation/demo

Evaluation

Projects will be evaluated based on:

- Security correctness
- Faithfulness to selected password manager
- Quality of entropy and randomness implementation
- Interface usability
- Code clarity